JOB PORTAL MANAGEMENT SYSTEM

CASE STUDY REPORT

Submitted by: _____

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Project Focus: Recruitment workflow, requirements analysis, database design, and evaluation

1. Abstract

The Job Portal Management System is a case study of how a simple digital platform can organize basic recruitment activities for job seekers and employers. Job seekers can create an account, explore available vacancies, search for relevant opportunities, submit applications, and review application progress. Employers can publish vacancies, view applicants, and update application status. The system is designed as a local, menu-driven application, with records stored in an SQLite database.

The case study examines the problem, stakeholders, requirements, system design, data organization, security considerations, testing approach, limitations, and possible future improvements. It demonstrates how a small information system can reduce scattered records and make recruitment information easier to manage.

2. Introduction

Recruitment involves several connected activities: publishing vacancies, collecting applications, reviewing candidate details, and communicating decisions. When these activities are handled through separate emails, messages, and spreadsheets, information can become difficult to organize. Candidates may lose track of the jobs they have applied for, while employers may find it challenging to maintain a clear record of applicants and decisions.

A job portal provides a centralized place to organize these activities. This project focuses on the essential functions of a job portal rather than the complexity of a commercial recruitment website. It is intended as an academic case study that demonstrates problem analysis, requirements gathering, database design, user roles, and basic information management.

3. Problem Statement

The main problem addressed is the lack of a single, organized record for job listings and applications in a small recruitment workflow. Manual tracking may result in duplicated information, difficulty finding relevant vacancies, uncertainty about application progress, and inconsistent status updates.

The proposed system provides a structured workflow in which job listings and applications are stored centrally. Users can access functions according to their role, and application information can be reviewed without relying entirely on separate documents or message threads.

4. Objectives

• Create a simple system that supports the basic activities of job seekers and employers.

• Allow users to register and sign in under an appropriate role.

• Provide browsing and keyword-based searching of job vacancies.

• Allow job seekers to submit applications and review their status.

• Allow employers to post vacancies, review applicants, and update application status.

• Store user, job, and application records persistently in a database.

• Apply input validation and basic password protection.

• Identify limitations and recommend realistic future improvements.

5. Scope of the Study

The project covers a local prototype of a job portal. Its scope includes account management, role-based menus, vacancy management, job searching, application submission, application tracking, and employer review of applications. Application details can include a resume reference or portfolio link and a cover letter.

The system is intended for demonstration and learning. It is not a fully deployed recruitment service and does not provide a public website, direct resume-file hosting, automated email notifications, online payment, or advanced recruitment analytics. These features are outside the current scope and are considered potential future enhancements.

6. Stakeholders and Their Roles

Job seekers use the system to browse opportunities, search for relevant vacancies, apply for jobs, and check application progress. Their main need is to find suitable opportunities and maintain a clear view of submitted applications.

Employers use the system to publish job openings, review the vacancies they have posted, inspect applications received, and record decisions or progress updates. Their main need is to organize recruitment information in a consistent way.

The project maintainer is responsible for maintaining the application, checking that the database works correctly, improving validation, and addressing software issues. In a larger deployment, an administrator role could also manage user accounts, reports, and policy compliance.

7. Requirements Analysis

Functional requirements describe what the system should do. The system should support account registration and login, job browsing, keyword search, job posting by employers, application submission by job seekers, application-history viewing, applicant review, and application-status updates. It should prevent duplicate usernames and duplicate applications for the same job where applicable.

Non-functional requirements describe expected qualities. The interface should be understandable for a beginner, stored records should persist between sessions, invalid input should be handled appropriately, and the system should be maintainable and portable to environments with a compatible Python installation. Basic password protection and data integrity are also important.

8. Proposed System and Workflow

The system begins by preparing its database and displaying the available menu. A user selects an action, provides the required information, and the system validates the input before carrying out the operation. The result is then presented to the user, who can continue using another menu option or exit.

For job seekers, the typical workflow is registration or login, browsing or searching vacancies, selecting a suitable role, submitting an application, and checking application status later. For employers, the workflow is login, posting a vacancy, viewing posted jobs, reviewing applicants, and updating application status. Keeping these workflows separate helps users focus on the tasks relevant to their role.

9. Database Design

The system uses SQLite, a lightweight relational database suitable for a small local application. The database is organized around three main entities: users, jobs, and applications.

Users: stores account information, including a unique username, password-hash information, and role. The role distinguishes job seekers from employers.

Jobs: stores vacancy details such as job title, company or employer information, description, location, and the employer associated with the posting. Each job record has a unique identifier.

Applications: connects a job seeker to a job and stores application-related information such as a resume reference, cover letter, current status, and submission date. Relationships between these records make it possible to associate an application with both the candidate and the vacancy.

Separating these entities reduces unnecessary repetition and makes it easier to retrieve related information. Database constraints and parameterized SQL statements help maintain consistency and reduce avoidable data-entry errors.

10. Security and Data Integrity

Account protection is an important consideration even in an academic prototype. Passwords should not be stored as plain text. The project uses salted password hashing, which stores a derived value rather than the original password. SQL queries use placeholders for supplied values, reducing the risk of unsafe query construction.

Data integrity is supported by unique identifiers and constraints that help prevent duplicate usernames and duplicate applications. Input validation is also necessary to avoid incomplete or inappropriate records. These measures provide basic protection but do not make the prototype suitable for handling sensitive information at a public, production scale.

A production system would require additional controls, including secure deployment, stronger session management, account recovery, access logging, protection against abuse, privacy notices, backup procedures, and a review of applicable data-protection requirements.

11. Testing and Evaluation

Testing should verify both expected actions and invalid cases. Registration should be checked with valid details and with a username that already exists. Job posting should be checked to confirm that a new vacancy is stored and appears in the listings. Search should return matching vacancies when suitable keywords are entered.

Application submission should be tested using a valid job and candidate account. A repeated application for the same vacancy should be rejected or otherwise handled according to the system's duplicate-prevention rule. Application tracking should show the saved status, and employer status updates should remain available when the relevant record is viewed again.

These checks evaluate the main workflow and help identify errors in validation, database relationships, and status management. Further testing could include empty fields, unusually long input, invalid login details, and database recovery scenarios.

12. Benefits and Critical Analysis

The main benefit is centralization: job details and application records are kept in one place instead of being spread across multiple documents. Keyword search helps users locate vacancies, while application tracking provides a clearer view of progress. Separate role-based menus make the workflow easier to understand, and persistent database storage allows records to remain available between sessions.

However, the current design has important limitations. A command-line interface is less accessible and visually engaging than a modern website. Resume information is represented by a reference or link rather than a managed upload. The system does not send notifications, provide advanced filtering, or support large numbers of concurrent users. It also lacks the administrative and operational controls expected of a real recruitment platform.

Therefore, the system should be viewed as a learning prototype that demonstrates core concepts, not as a replacement for a commercial job portal.

13. Future Enhancements

Future development could introduce a web interface using a Python framework such as Flask or Django. A responsive interface would make the portal easier to use across desktop and mobile devices. Secure resume uploads, email notifications, password reset, and profile management could improve the experience for both candidates and employers.

Additional improvements could include filters for location, job type, experience level, and skills; employer dashboards; administrative tools; audit logs; stronger authorization controls; automated backups; and deployment to a managed server. Before a public launch, the system would also need broader security testing, accessibility checks, privacy

safeguards, and performance evaluation.

14. Conclusion

The Job Portal Management System addresses a common organizational problem: keeping vacancy and application information clear and accessible. By supporting separate job-seeker and employer workflows and storing records in a relational database, the project demonstrates how a small information system can structure recruitment tasks.

The case study highlights the importance of requirements analysis, database design, input validation, basic security, and testing. Although the current prototype has limitations in interface design, notifications, file handling, and deployment, it provides a foundation that can be extended into a more complete web-based recruitment system.